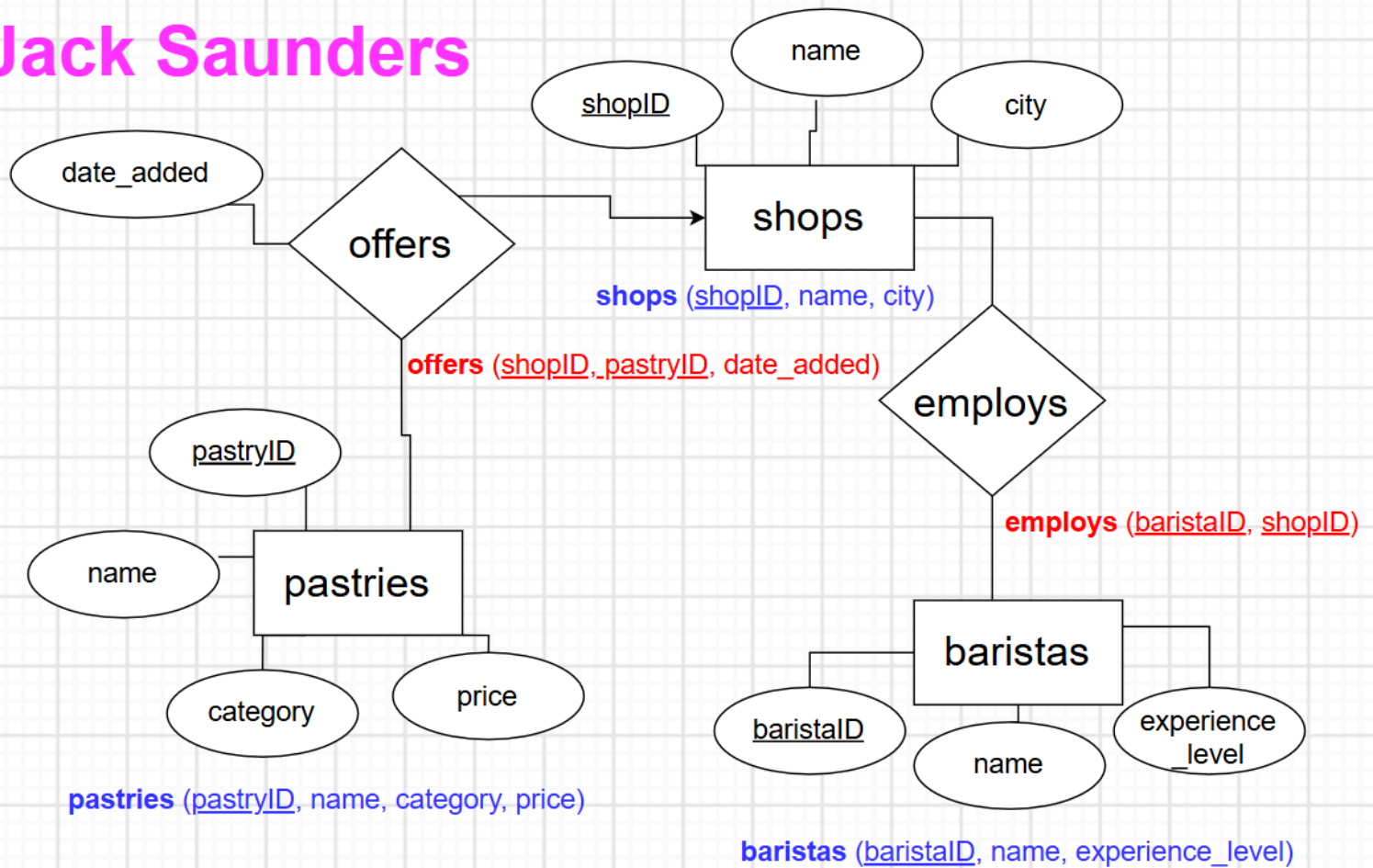


Title: Database Assignment 2

Your Name: Jack Saunders

Date: 9/22/2026

Jack Saunders



Assumptions:

A) **offers** is a one-to-many relationship due to each **shop** offering two different **pastries**

B) How can any **barista** get to work on time?!?! They are working across America according to **employs**! They must be teleporting!

1. Find the average price of pastries for each category from the pastries table

```
Select category, avg(price)
From pastries
Group by category;
```

```
13 -- 1) Find the average price
14 • Select category, avg(price)
15 From pastries
16 Group by category;
17
```

Result Grid			Filter Rows:		Export
	category	avg(price)			
▶	Pastry	4.666666666666667			
	Muffin	3.625			
	Bakery	4.25			
	Scone	3.85			
	Breakfast	8			

2. Find the total number of baristas at each experience level from the baristas table

```
Select experience_level, count(*)
as "# baristas"
From baristas
Group by experience_level;
```

```
18 -- 2) Find the total number of baristas at each e
19 • Select experience_level, count(*) as "# baristas"
20 From baristas
21 Group by experience_level;
22
```

Result Grid			Filter Rows:		Export:	Wrap Cell Content:
	experience_level	# baristas				
▶	Junior	2				
	Senior	2				
	Mid-Level	2				

3. Count the total number of shops located in each city from the shops table

Select city, count(*) as "# shops"
From shops
Group by city;

```
23 -- 3) Count the total number of sho
24 • Select city, count(*) as "# shops"
25 From shops
26 Group by city;
```

Result Grid		Filter Rows:	Export:	Wri
city	# shops			
Seattle	1			
Portland	1			
Austin	1			
Chicago	1			
Denver	1			
Boston	1			

4. Find the maximum price among pastries for each category from the pastries table

Select category, max(price)
From pastries
group by category;

```
28 -- 4) Find the maximum price
29 • Select category, max(price)
30 From pastries
31 group by category;
32
```

Result Grid		Filter Rows:	Exp
category	max(price)		
Pastry	5.5		
Muffin	3.75		
Bakery	4.25		
Scone	3.85		
Breakfast	8.5		
Cake	4		

5. Count how many pastries have been added by each shop using the shopID column from the offers table

Select shopID as "shop", count(shopID)
as "# pastries"
From offers
Group by shopID;

```
33 -- 5) Count how many pastries have been added by each
34 • Select shopID as "shop", count(shopID) as "# pastries"
35 From offers
36 Group by shopID;
37
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: 1A

	"shop"	# pastries
1		2
2		2
3		2
4		2
5		2
6		2

6. Find the name, category, and price of any pastry whose price matches the maximum price within its category

Select name, price, category
From pastries p
Where price =
(Select max(price)
From pastries c
Where p.category = c.category);

```
38 -- 6) Find the name, category, and price
39 • Select name, price, category
40 From pastries p
41 Where price =
42 (Select max(price)
43 From pastries c
44 Where p.category = c.category);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: 1A

	name	price	category
1	Almond Croissant	5.5	Pastry
2	Blueberry Muffin	3.75	Muffin
3	Cinnamon Roll	4.25	Bakery
4	Chocolate Scone	3.85	Scone
5	Blueberry Scone	3.85	Scone
6	Breakfast Burrito	8.5	Breakfast
7	Lemon Loaf	4	Cake

7. Find the unique shop IDs from the offers table that have offered at least one pastry whose price is strictly greater than the overall average price of all pastries

```
Select distinct shopID
From offers
Where pastryID in
  (Select pastryID
   From pastries
   Where price >
    (Select avg(price)
     From pastries));
```

```
46  -- 7) Find the unique shop
47 •  Select distinct shopID
48  From offers
49  Where pastryID in
50      (Select pastryID
51      From pastries
52      Where price >
53          (Select avg(price)
54          From pastries));
```

Result Grid			Filter Rows:		Export:
shopID					
▶ 1					
5					

8. Find the shop ID and pastry ID for the records in the offers table that have the earliest **date_added** (minimum date)

```
Select *
From offers
Where date_added <= all
  (Select date_added
   From offers);
```

```
56  -- 8) Find the shop ID a
57 •  Select *
58  From offers
59  where date_added <= all
60      (select date_added
61      From offers);
```

Result Grid			Filter Rows:		Exp
shopID pastryID date_added					
▶ 1 1 1/10/2026					

9. Find the shop ID(s) that offer the highest number of pastries, utilizing a subquery to evaluate the maximum count per shop

```
Select shopID
From offers
Group by shopID
Having count(pastryID) >= all
      (Select count(shopID)
       From offers
       Group by shopID);
```

```
63  -- 9) Find the shop ID(s) that
64 • Select shopID
65  From offers
66  Group by shopID
67  Having count(pastryID) >= all
68      (Select count(shopID)
69      From offers
70      Group by shopID);
```

Result Grid		Filter Rows:	Export:	Wi
	shopID			
▶	1			
	2			
	3			
	4			
	5			
	6			

10. Find the names of baristas who work at shops located in 'Seattle' using nested subqueries

```
Select name
From baristas
Where baristaID in
      (Select baristaID
       From employs
       Where shopID in
            (Select shopID
             From shops
             Where city = 'Seattle'));
```

```
72  -- 10) Find the names of baristas
73 • Select name
74  From baristas
75  Where baristaID in
76      (Select baristaID
77      From employs
78      Where shopID in
79          (Select shopID
80          From shops
81          Where city = 'Seattle'));
```

Result Grid		Filter Rows:	Export:	Wrap C
	name			
▶	Emma Watson			
	Liam Chen			